# Tsz Fei Wang

Kitchener, Waterloo | wangtszfei@gmail.com | 647-939-3797 | LinkedIn | GitHub | Portfolio

## Education

**University of Waterloo** | Honours Computer Science (Co-op) — Sept 2026 – May 2031 (expected)

## Skills & Technologies

**Languages & Frameworks**: Python, SQL, R, Java, C++, TypeScript
**Tools**: PyTorch, NumPy, Pandas, FastAPI, Jupyter, Kaggle, Elasticsearch, Gemini API, Git, Linux, Docker, GitHub Actions

## Experience

**Century Frontier Asset Management** — Jul 2026 – Aug 2026
*Quantitative Research Intern* — *Shenzhen, CN*

- Built a point-in-time backtesting pipeline in **Python** and **pandas** that eliminated lookahead bias from the firm's LLM-driven equity research across **104 tickers**.
- Benchmarked **9 LLM architectures** on stock-preference and inter-stock-relationship prediction, showing that a model's stated confidence was a poor predictor of its accuracy.

**Krembil Research Institute, World Community Grid** — Mar 2026 – Jun 2026
*Software Engineer Intern* — *Toronto, ON*

- Built most of the frontend for Catrin's full-stack data access portal, including the search, filter and results views researchers use to explore **~4 million** TF-TG associations across **33,000 genes**.
- Reconciled the schemas of **9 independent data sources** so the portal can query them as one dataset.
- Listed as a contributor on the Catrin manuscript (in preparation).

**Non-Trivial Research Fellowship** — Mar 2025 – Aug 2025
*Machine Learning Research Fellow* — *Remote*

- Trained and evaluated a **PyTorch** audio deepfake classifier in **Python** on a Kaggle dataset of real and AI-synthesized speech.
- Researched AI voice cloning as an emerging security threat and proposed a detection-based defence in a written research proposal.

**Mackenzie Cybersecurity Team** — Jun 2025 – Jun 2026
*Team Captain, President* — *Toronto, ON*

- Captained the team to **First in Canada** at CyberTitan, Canada's largest cyberdefense competition, hardening Windows, Linux and Cisco systems under time pressure.
- Created and taught a cybersecurity curriculum reaching **60+ students** across the school.

## Projects

**Technova Hackathon Badge** | *KiCad, Hardware Lead* — *Oct 2026*

- Led hardware design of a custom NFC badge for a **~300-person** hackathon in **KiCad**, taking it from schematic to a manufacturing-ready JLCPCB assembly order.
- Cut the per-badge cost from about **$11 to $7** by reworking the parts list around in-stock LCSC components, while fitting NFC check-in, an OLED display, LEDs and USB-C charging.

**Submerge** | *GLSL, QML, Python* — *Oct 2026 – Present*

- Shipped an animated KDE Plasma theme in **GLSL** and **QML** downloaded **~1,290 times** on the KDE Store, with a shader water simulation and a live system HUD.
- Resolved a user-reported NVIDIA/Wayland rendering failure by porting the shaders to GLSL ES, and added power-aware frame pacing to save battery.

**Physical Studio** | *Python, React, Three.js, Docker* | ***Winner**, Hack the Valley 11* — *Oct 2026*

- Won Hack the Valley 11 with a tool that turns a prompt into a step-by-step hardware build guide; engineered the **Python** MCP server that lets an LLM design from measured parts and check its own work for collisions.
- Built the **React Three Fiber** viewer that animates each assembly step and deployed the app with **Docker Compose** on Vultr.

## Awards & Honors

**National Champion x2**, 1st in Canada, Cybersecurity, $1,000 scholarship — May 2025, Mar 2026
**National Finalist**, 2nd in Canada, Cybersecurity, $1,600 scholarship — Mar 2025